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CSE461 Sec 8 Project Report Submission

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BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *Smart Autonomous Hospital Delivery & Safety Monitor Robot*

by

Group_No-3
Section-8

Department of Computer Science and Engineering
BRAC University
DATE-04/09/2026

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Abstract

The project is titled Smart Autonomous Hospital Delivery & Safety Monitor Robot, focusing primarily on delivering healthcare supplies, medications, testing materials, and health records while connecting dispensary, inpatient units, diagnostic centers, and numerous other departments, hence significantly minimizing the amount of work for nurses.The Arduino Uno R3, which serves as the robot's brain, is connected to a 5-channel infrared sensor to help locate and track the floor line, an ultrasonic sensor to sense objects in its path, an MFRC522 RFID reader to identify intersections and target locations, flame and smoke sensors to detect signs of flames or smoke, a servo motor to control access to the medicine box, a buzzer to issue audible alerts, a status LED to indicate system condition, and a 16x2 I2C LCD displays information and alerts.Essentially, this self-operating device was created to include various operational modes such as moving, turning, halting, and delivering. Alongside the system, several tests were conducted on a simulated hospital route where we faced challenges such as path detection, faulty parts, and accurate turning, but ultimately, all were resolved.Once all evaluations of the robot were completed, the conclusion indicated that the system could effectively identify flames and smoke, navigate using black line and RFID, and finally detect an object within a 20cm range and classify it as a block. The system proves to be eco-friendly by being reusable, rechargeable and repairable.

Keywords

Arduino Uno, RFID navigation, autonomous hospital delivery, obstacle avoidance, finite-state machine.

1. Introduction

Healthcare human services require time to transfer medicinal products, clinical samples, examination results, and more, which ultimately seizes patients in receiving healthcare services. Additionally, there are instances when medications require careful oversight to be transported securely without being damaged, a service that human resources might offer less, and ultimately, the delay would cease to be an issue.To address all these challenges, we have created this robot to effectively determine a path and self-locate using RFID, also to tackle the issue of damaging the medications, checking for obstacles ahead, and finally to sense smoke or flames.The system has been assessed on a path that features a defined route, including a pharmaceutical unit, a research facility, and a care unit as three wards. In addition, it is closely related to Computer Interfacing since the Arduino Uno microcontroller manages various components such as sensors, motor drivers, servos, LCDs, and others. The initiative employs a State-based navigation approach for aligning the components and managing the autonomous device.

2. Related Work/ Inspiration

The constructed autonomous system is not novel; similar systems are already implemented in latest, high-cost environments for transporting medicines. These systems utilize depth cameras rather than ultrasonic sensors for object localization. Additionally, cost-effective versions of the lock and unlock mechanism exist, and the system relies on specialized software for multi-robot management.Our solution primarily combines the most recent characteristics for such robots with their cost. For example, unlike simple robots that can follow straight pathways, we provided an appropriate technique to deliver the package in accordance with the user-selected destination in addition to basic tracking. We included a security mechanism that only allows access once it has reached its intended location in order to ensure safe handling. Additionally, this robot provides an affordable service and can quickly identify impediments that are located within 20 cm of our system.

3. Technical Approach

8 First, the Arduino Uno R3 serves as the system's central controller, controlling every other module. Additionally, the use of all the sensors aids in delivering an output through a state-driven approach. Besides, they utilize a minimal voltage of 5V, which can be easily supplied by the Arduino. However, the motors require an elevated voltage, so we integrated the L298N component to drive them while ensuring the Arduino remains operational without restarting. The block diagram shows the hardware linkages, whereas the flowchart depicts the operating processes of the automated system.

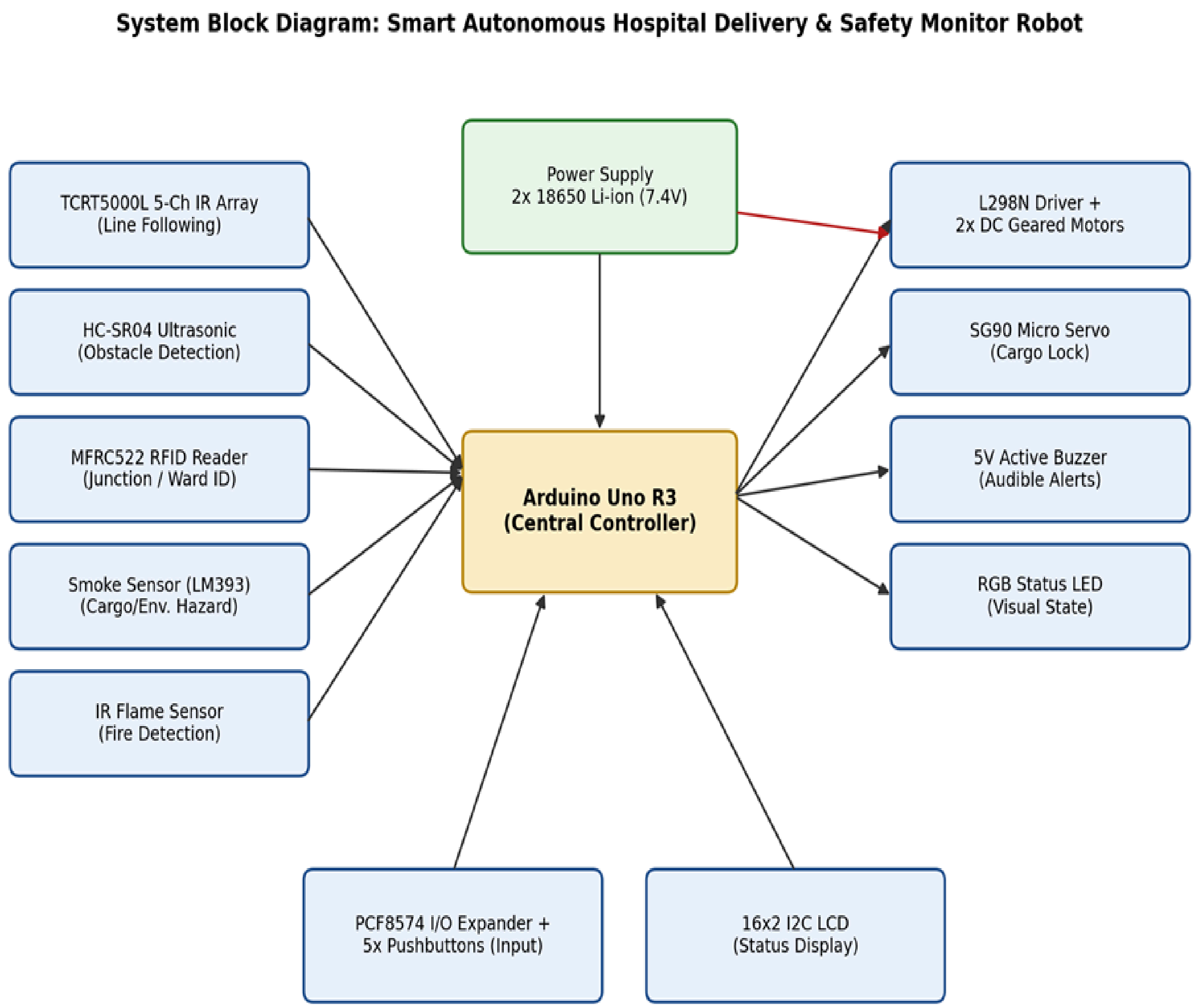

Figure 1: Block diagram of the autonomous system

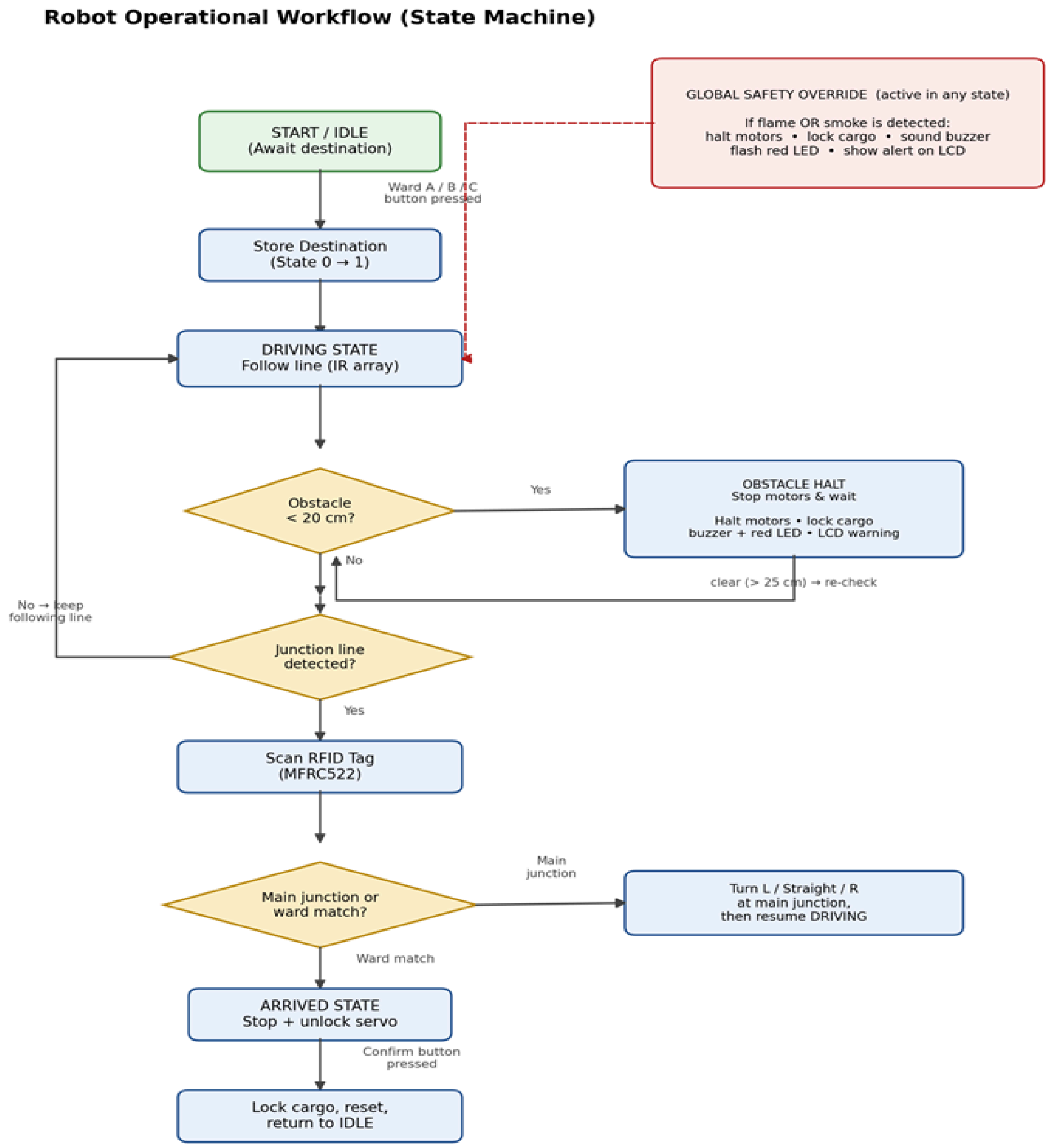

Figure 2 : Flowchart of the autonomous system

3.1 Components Used:

The system's hardware is all listed in the Table below. The automated machine was developed using Arduino from a programming standpoint, utilizing C as the language and coded within the Arduino IDE. The system operated using the FSM approach, consisting of 5 phases that continuously cycle within Arduino's main loop function. Specialized libraries such as MFRC522, Servo, Wire (I2C), and LiquidCrystal_I2C facilitate interaction with the RFID reader, locking system, and PCF8574/LCD components.

Category	**Components**	**Qantity**	**Purpose**
Microcontroller	Arduino Uno R3	1	Centralized system management
Locomotion	2WD Single-Layer Acrylic Chassis (incl. 2x 5V DC geared motors, caster wheel)	1	Robot chassis and motion system
Locomotion	L298N Dual H-Bridge Motor Driver	1	Controls dual DC motors
Sensor	TCRT5000L 5-Channel IR Line Array	1	Path-tracking system
Sensor	HC-SR04 Ultrasonic Sensor	1	Obstruction sensing
Sensor	MFRC522 RFID Reader + tags	1	Target location recognition
Sensor	Smoke Sensor (MQ-2, replaced DHT11)	1	Surroundings l risk identification
Sensor	IR Flame Sensor	1	flame recognition
Actuator	SG90 Micro Servo Motor	1	Storage access control
Actuator	5V Active Buzzer	1	Audio warnings

Actuator	Common Cathode LED	1	Display indicators
UI / Logic	16x2 LCD with I2C backpack	1	Information and target screen
UI / Logic	PCF8574 I2C I/O Expander	1	Interface connection extension
UI / Logic	Tactile pushbuttons	4	Target choice and approval
Power	18650 Li-ion battery (3.7V)	4(2-2 parallel)	Energy source
Power	Dual-slot 18650 battery holder	2	Energy pack holder
Accessory	Breadboard, standoffs, jumper wires	According to requirement	Model building

Table 1: Components needed

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3.2 Cost Breakdown:

Components	Unit Cost	Quantity	Total
Arduino Uno R3	950	1	950
2WD Acrylic Car Chassis Set (incl. DC motors/caster)	570	1	570
L298N H-Bridge Motor Driver	185	1	185
MFRC522 RFID Module with Key Fob/Card Tags	190	1	190
16x2 LCD Display with integrated I2C Backpack	255	1	255
PCF8574 I2C I/O Expansion Module	160	1	160

Breakout			
TCRT5000L 5-Channel IR Line Tracking Array	310	1	310
HC-SR04 Ultrasonic Sensor Module	130	1	130
Smoke / DHT11 Sensor Module	120	1	120
IR Flame Sensor	90	1	90
SG90 Micro Servo Motor	130	1	130
18650 Rechargeable Li-ion Batteries	260	2	520
Misc. (Buzzer, Pushbuttons, LEDs, standoffs, wires, Tape, Breadboard)	450	1	450
Total			4060

Table 2 : Cost Breakdown

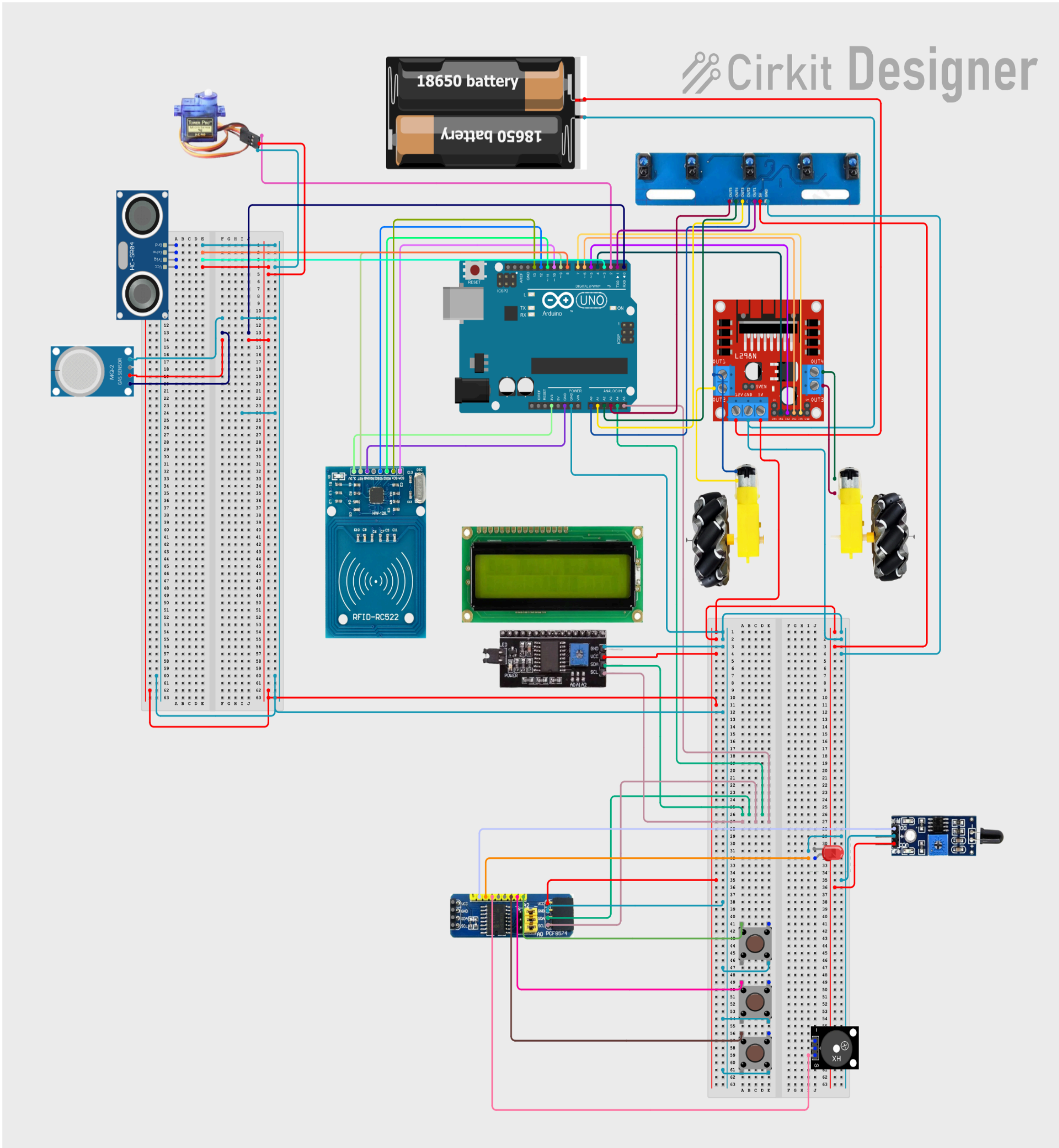

Figure 3 : Circuit/Schematic diagram

3.3 Functionality

Step1: The system begins from its designated starting point and remains ready until a request is received.

Step 2:A button is pressed to choose the destination (Ward A/B/C), and the PCF8574 assists in linking these buttons to the Arduino.

Step 3: Once the destination is chosen, the robot begins to move and utilizes infrared sensors to effectively follow the path, while constantly identifying barriers under 20 cm using the HC-SR04 ultrasonic sensor.

Step 4: Upon reaching the Primary junction, the system employs the RFID and ultimately decides on the chosen destination (Left/Straight/Right) that was entered in step 1.

Step 5: The infrared sensor array detects the end of line by sensing white floor on all its IR sensors. Once the end of line is detected, the RFID sensor is activated to scan the destination card.

Step 6: After successful authentication of the correct destination, the robot provides access to the package compartment by activating the SG90 servo motor which unlocks it. Once the package has been retrieved, and the confirm button is pressed, the compartment is secured again.

Step 7: This may occur at any moment, regardless of what the robot is doing at that time.If the flame sensor or smoke sensor senses danger, the emergency condition takes precedence, it will instantly halt and switch to emergency mode.

3.4 Challenges:

Initially, the arduino could not establish a connection with RFID however an unstable module connection was identified. Soldering and securing the module to the circuit board resolved the issue. We started with only one pair of batteries but the capacity was not enough for longer hours of testing and usage so we upgraded the power system and added another pair of batteries to up the cumulative capacity. Due to the numerous sensors, buttons and actuators, many input and output pins were needed, so pins like buttons and LEDs were transferred to the PCF8574 I2C expander. Determining how to monitor the thermal sensors was challenging, and during this process, the sensor malfunctioned and the faulty unit was exchanged for a backup digital smoke sensor which worked perfectly. Finally, the robot's weight was not balanced due to the heavy battery packs so we had to tune the power supply of each motor driver separately for flawless turning and forward movement.

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4. Sustainability & Impact

4.1 Sustainability:

The system is an extremely sustainable unit due to its use of Li-ion batteries, which have the advantage of a longer shelf life than standard batteries, as they can be recharged once depleted. In addition, the chassis is well designed with body components that can be dismantled and utilized in various systems. Additionally, since our project is entirely based on being cost-efficient, it is constructed from readily accessible components, allowing for repairs to occur without concern.

4.2 Impact:

Typically, nurses need to handle report transfers, medication, and testing supplies. However, the implementation of these automated machines will enable nurses to focus on tasks that are more beneficial for the hospital and dedicate more time to patient care. Consequently, the system will reduce human to human communication which can cause delays, leading to better clinical care for the patients. Our system is highly cost-effective and suitable for developing countries or underdeveloped regions where hospitals struggle to hire sufficient staff for minor tasks. Ultimately, this also aids in reducing risks for the hospital by identifying smoke, dangerous gases and flame.

4.3 Future Work:

Currently, our system has some future improvements as it lacks a temperature sensor. Instead, it relies on a smoke sensor. This could be significantly improved if both sensors were combined, with one measuring temperature and the other assisting in smoke detection. Additionally, an issue occurs with the power system, the batteries are rechargeable but requires human intervention for charging. However, if an automatic docking and charging system could be incorporated, the process would be significantly more seamless.

4.4 Limitations:

The Arduino mainly has a limited number of input and output pins, leading to a limited set of modules that can be used at a time. Secondly, an RGB LED instead of a RED LED could enable various colors to represent different states. Additionally, the integrated path is a predetermined route with specific target points, while the real-world path is significantly more intricate and complicated. Furthermore, the change in direction lacks precision as it is greatly influenced by time. Every addition of a different kind of turn, which is not a strict 90 degrees, would require manual tuning in the code to perfect it. Once the batteries are low on power, the robot effectiveness is compromised. Thus, an additional module to indicate the power level of the batteries would benefit users.

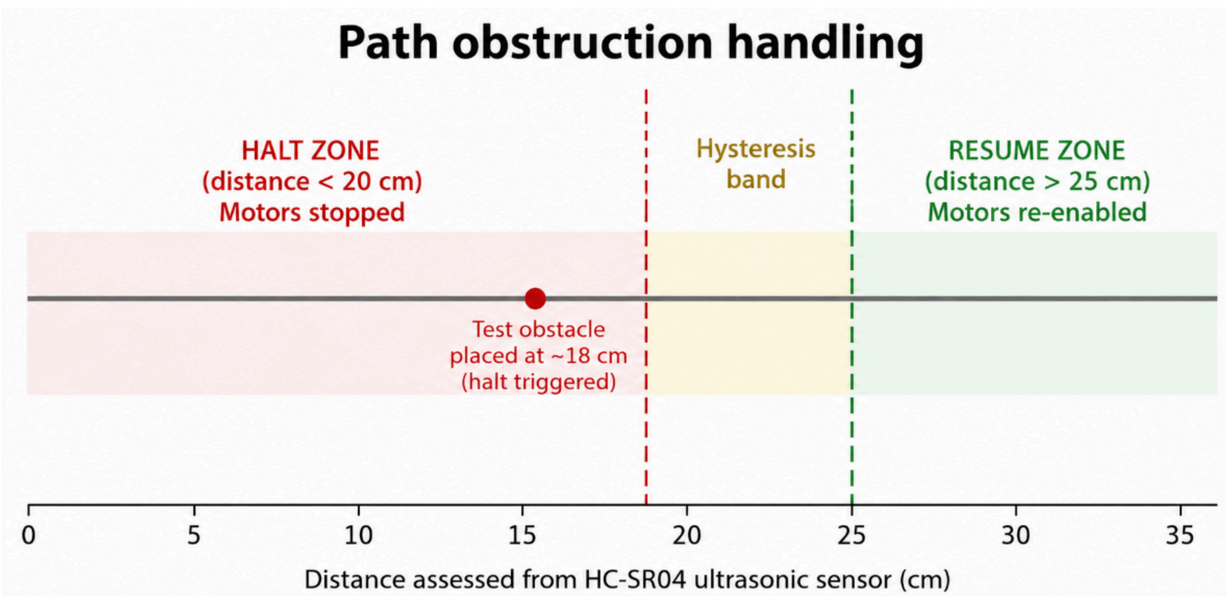
5. Results & Discussion

Over the course of testing, the object localization functioned flawlessly with the ultrasonic sensors when a hand was maintained at a distance of less than 20 cm, resulting in clear detection. In addition to the system's evaluation on certain positions, which produced well-calibrated output, there was a little temporal discrepancy that prevented it from twisting to the appropriate angle; hence, a small time lag was introduced to make the machine function properly. Last but not least, we used a flame stick to get data about any ignition source and fumes to monitor any emissions.

Specification	Progress	Remarks
Route target input via pushbuttons (PCF8574)	Achieved	The PCF8574 provides additional input channels for five buttons used to select the target and verify delivery.
Self-guided path tracking (5-channel IR array)	Achieved	Consistent movement along marked routes and accurate intersection recognition after adjusting the sensor position.
Operational information display (LCD, buzzer, LED)	Achieved	The 16×2 I2C display and audible alert operated correctly; a single warning light was used instead of the planned RGB signal.
Path obstruction sensing (HC-SR04)	Achieved	The system pauses near objects at a 20 cm distance and resumes travel when no blockage remains.

RFID-assisted navigation and target location detection (MFRC522)	Achieved	Operated through the FSM; communication was restored by securing the module to the board.
Flame sensing capability (IR sensor)	Achieved	Managed by a central warning system that can interrupt any operating mode.
Compartment security mechanism (SG90 servo)	Achieved	Remains secured while traveling; access is granted after the RFID system verifies the destination.

Table 3:Comparison between the initial design and final system



6. Conclusion

Our project aims to achieve extraordinary success in benefiting hospitals, with objectives centered on financial practicality and improving healthcare efficiency.

Consequently, this self-sufficient device can carry out multiple tasks such as detecting smoke and flame while locating destinations using RFID to reach the intended target. It can also respond to obstacles when they arrive. Additionally, locking the medicines until the robot arrives at its destination is an important capability. Nonetheless, there were some constraints in developing this project, such as the temperature sensor malfunctioning, the RFID voltage delivery issue, and the RGB LED not being used. Additionally, despite its numerous shortcomings, the autonomous machine continues to have its own practical benefits. For example, it is very easy to maintain because all of its components are easily sourced, and it is more financially practical for hospitals to have it rather than hiring a human to perform such menial tasks.

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7. Contribution

Name	ID	Role	Specific Contribution
Md. Rafiul Islam	22201842	Circuit Maker & Technical Coordinator	• Spearheaded the actual project build, authoring the core Arduino firmware, state-machine logic, and PID line-tracking implementation. • Collaborated on compiling technical specifications and system architecture details for the project report and documentation.
Lamisa Ibrahim	23101132	Documentation & System Tester	• Led the report-making process by authoring primary chapters, component specifications, and cost breakdowns. • Participated in initial hardware testing phases, functional verification, and use-case validation of the delivery robot.

Sabrina Jahan	22101034	Visual Designer & Testing	• Designed the project poster, visual layouts, system diagrams, and presentation hierarchy for the final showcase. • Helped test navigation responses, UI pushbuttons, and display telemetry during physical integration runs.
Md Nazim Hossain	22299256	Hardware Assistant & Debugging	• Assisted with physical assembly, chassis wiring, and hands-on hardware testing and troubleshooting on the test track. • Contributed to gathering data, structuring content, reviewing and rewriting technical sections for the final project report.
Moriom Binta Abdul Malek	23301359	Operator controls with guidance	• Co-authored and formatted the project report, detailing system functionalities, workflows, and sensor integration. • Assisted with circuit verification, debugging workflow checks, and assembling physical components during testing.

References

4 Arduino. (n.d.). *Arduino Uno Rev3 documentation*. Arduino Docs. https://docs.arduino.cc/hardware/uno-rev3

5 Balboa, M. (2012). *rfid: Arduino RFID library for MFRC522* [Computer software]. GitHub. https://github.com/miguelbalboa/rfid

1 Components101. (n.d.). *HC-SR04 ultrasonic sensor: Working, pinout & datasheet*. https://components101.com/sensors/ultrasonic-sensor-working-pinout-datasheet

6 Components101. (n.d.). *DHT11 temperature and humidity sensor module*. https://components101.com/sensors/dht11-temperature-sensor

3 Texas Instruments. (2000). *L298 dual full-bridge driver* (Datasheet). https://www.ti.com/lit/ds/symlink/l298.pdf